

Logarithmic boundary depth and the conditioning of the discrete inverse conductance problem on hyperbolic lattices

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Abstract

We study the discrete inverse conductance (Calderón) problem—recovering the edge conductances of a resistor network from boundary voltage–current data—on layer-truncated hyperbolic $\{7, 3\}$ tilings, and compare it with square and triangular lattice disks of matched node count N . Using an exact formula for the sensitivity Jacobian of the Dirichlet-to-Neumann (DtN) map, we find that its condition number κ grows polynomially on the hyperbolic lattice ($\log_{10} \kappa = 3.89$ at $N = 847$, inside a preregistered prediction band), while on the flat lattices it reaches $10^{9.7}$ at $N = 317$ and exceeds the resolution of IEEE double precision by $N \approx 400$; in 512-bit ball arithmetic it continues as $e^{c\sqrt{N}}$, reaching $10^{16.5}$ at $N = 797$, within 0.2 decades of the value extrapolated from the unsaturated data and far from any power law. The mechanism is geometric: the maximal distance from a node to the boundary grows as $O(\log N)$ on the hyperbolic lattice and as $O(\sqrt{N})$ on flat ones; across five hyperbolic tilings $\log \kappa$ is nearly a function of $d_{\max}$ alone (spread at most 0.56 decades at equal depth), whereas at equal depth flat lattices are at least 1.8 decades worse, because the dependence on depth is concave for hyperbolic tilings and linear for flat ones. Exact rank computations over two finite fields show that every tested network is algebraically identifiable; the difference is one of conditioning, not of identifiability. The advantage holds for the Neumann-to-Dirichlet map measured by current-injection hardware and survives inhomogeneous conductances (uniform disorder, two-decade log-uniform disorder, and a $\times 100$ defect); at matched probe count *and* matched dimension, however, it shrinks to at most half a decade, which corrects a reading in version 1.0. For the time-resolved RC version of the network, a preregistered finite-size prediction that the Dirichlet spectral gap is constant was refuted; we show instead, by domain monotonicity and the positive spectral gap of the nonamenable $\{7, 3\}$ graph, that the gap decreases monotonically to a strictly positive limit, so the RC relaxation time is bounded uniformly in N , whereas it grows linearly in N on flat lattices. The RC dynamics are cross-validated with two independent BDF integrators (SciPy and the rusty-SUNDIALS CVODE solver), both of which reproduce the exact matrix-exponential solution and the static Dirichlet-to-Neumann map to within 3×10^{-8} and 10^{-11} respectively. In simulation, a single-node defect is detected and localised at the 3×10^{-4} precision budget on both geometries, but the flat lattice fails at measurement noise $100\times$ below the hyperbolic one at $N \approx 316$, and component tolerance up to 5% does not degrade localisation. All data, code and a claim-by-claim verification ledger are released.

1 Introduction

Whether data measured on the boundary of a system determine its interior is a question that recurs across physics: in the bulk-boundary correspondence of topological phases [1–3], and in holographic duality, where boundary entanglement data encode bulk geometry [4, 5]. Its simplest classical instance is the inverse conductivity problem on a resistor network: given the map from

boundary voltages to boundary currents, recover the interior conductances, and quantify how measurement noise is amplified in doing so.

For planar networks with boundary nodes on the outer face (*circular planar* networks), identifiability is characterised combinatorially: a network is recoverable from its response matrix if and only if it is critical [6–8]. Stability is a separate matter. Resistor-network approaches to electrical impedance tomography (EIT) show that reconstructions become unstable as networks are refined [9–11], consistent with logarithmic stability estimates for the continuum and discrete Calderón problems [12].

Hyperbolic lattices—regular tessellations of the hyperbolic plane—have been realised in circuit quantum electrodynamics [13] and on printed circuit boards [14], and are used as discrete models of holography [15, 16]. Their defining metric property for the present purpose is that the boundary of a truncated tiling contains a fixed positive fraction of its nodes, so every node lies within $O(\log N)$ steps of the boundary. We ask whether this translates into a better-conditioned inverse problem, and answer quantitatively, with preregistered predictions, exact certificates where floating point is insufficient, and a time-domain formulation suited to a table-top measurement.

We did not find prior work stating conditioning results for inverse conductance problems on hyperbolic lattices, within targeted searches of the EIT, resistor-network and hyperbolic-lattice literatures; this is a statement about those searches, not a claim of priority.

2 Model and methods

2.1 Networks

Layer-truncated $\{7, 3\}$ tilings (heptagons, three per vertex) are generated in the Poincaré disk by reflecting the central heptagon across its edges, breadth-first by tile layer $L = 1, \dots, 6$ ($N = 35$ to 5887). Square and triangular lattice disks of radius R provide the flat comparison. Faces are constructed explicitly, and a node is a boundary node if and only if it is incident to the outer face; this is necessary because full-degree nodes can touch the outer face on both lattice families. Euler’s relation $V - E + F = 1$ is checked for every instance. All conductances are $g_e = 1$ unless stated.

2.2 Dirichlet-to-Neumann map and its sensitivity

With $L = \sum_e g_e (\delta_a - \delta_b)(\delta_a - \delta_b)^\top$ the weighted graph Laplacian and nodes partitioned into boundary (b) and interior (i), the DtN map is the Schur complement

$$\Lambda = L_{bb} - L_{bi} L_{ii}^{-1} L_{ib}. \quad (1)$$

Let H be the harmonic extension ($H_b = I$, $H_i = -L_{ii}^{-1} L_{ib}$), so $\Lambda = H^\top L H$. Since the harmonic extension minimises the Dirichlet energy for fixed boundary data, its variation does not contribute at first order, and

$$\frac{\partial \Lambda}{\partial g_e} = (h_a - h_b)(h_a - h_b)^\top, \quad e = (a, b), \quad (2)$$

where $h_v^\top$ is the row of H for node v . All edge sensitivities thus follow from one factorisation of L_{ii} ; eq. (2) agrees with central finite differences to 1.2×10^{-11} in our tests. The Jacobian J has one column per edge, containing the strictly upper-triangular entries of $\partial \Lambda / \partial g_e$ (the diagonal is determined by the zero row sums). We report $\kappa = \sigma_{\max}(J) / \sigma_{\min}(J)$, the factor by which relative noise in Λ can be amplified in a linearised reconstruction; a case is labelled numerically singular when $\sigma_{\min} \leq \varepsilon_{\text{mach}} \sigma_{\max}$.

2.3 Exact rank certificates

Floating-point rank decisions become unreliable precisely where κ is large. Because L has integer entries, for any prime p with $\det L_{ii} \not\equiv 0 \pmod{p}$ the harmonic extension and eq. (2) can be evaluated exactly in $\mathbb{F}_p$, and $\text{rank}_{\mathbb{Q}} J \geq \text{rank}_{\mathbb{F}_p}(J \bmod p)$. A full-rank result modulo p therefore certifies full rank over $\mathbb{Q}$; for rank-deficient cases we require agreement between $p = 2^{31} - 1$ and $p = 998244353$.

2.4 Condition numbers beyond double precision

Where $\kappa \gtrsim 10^{16}$, $\sigma_{\min}$ lies below the rounding error of J 's own entries and no double-precision SVD can recover it. We therefore compute κ for the double-precision-singular flat instances in ball arithmetic (Arb, through python-flint) at 512 bits, which tracks a rigorous error radius through every operation. The harmonic extension is solved from the integer Laplacian in Arb, and the Gram matrix $G = J^T J$ is formed in Arb without materialising J : with $d_e = h_a - h_b$ restricted to the boundary and rows of J indexed by boundary pairs $i < j$,

$$G_{ef} = \sum_{i < j} d_{e,i} d_{e,j} d_{f,i} d_{f,j} = \frac{1}{2} [(d_e \cdot d_f)^2 - (d_e^2 \cdot d_f^2)]. \quad (3)$$

G^{-1} is computed in Arb, and $\lambda_{\max}(G^{-1}) = 1/\sigma_{\min}^2$, a well-conditioned quantity, is read off after casting the certified midpoints to double precision; $\sigma_{\max}$ is read from G likewise. A value is reported only if every entry of G^{-1} has relative radius below 10^{-20} . As a control, the same code must reproduce the double-precision κ of two unsaturated instances to 10^{-6} in $\log_{10} \kappa$ before any saturated instance is reported. The rival forms for the flat growth law, and their extrapolations, were fixed in a preregistration before this computation was run.

2.5 Neumann-to-Dirichlet map

Hardware that injects currents and reads voltages measures the Neumann-to-Dirichlet map Λ^+ (the Moore–Penrose inverse, the inverse on the zero-sum subspace), with sensitivity $\partial \Lambda^+ / \partial g_e = -\Lambda^+ (\partial \Lambda / \partial g_e) \Lambda^+$.

2.6 RC network

Attaching a capacitance C from every node to ground and imposing boundary voltages V_b gives

$$C \dot{V}_i = -(L_{ii} V_i + L_{ib} V_b), \quad (4)$$

whose steady state is the harmonic extension and whose steady boundary currents are ΛV_b . The slowest relaxation time is $\tau = C/\lambda_{\min}(L_{ii})$ and the stiffness ratio is $\lambda_{\max}(L_{ii})/\lambda_{\min}(L_{ii})$. Equation (4) is the time-resolved signal a physical network sampled by a microcontroller produces.

2.7 Integrators and controls

Equation (4) is integrated with two independently developed implementations of variable-order BDF: SciPy's `solve_ivp` [17] and the CVODE solver of rusty-SUNDIALS [18], a Rust implementation of the SUNDIALS suite [19] exposed to Python through PyO3. Both use $\text{rtol} = 10^{-8}$, $\text{atol} = 10^{-10}$. Two controls gate every reported integration. **K1 (known answer)**: the solution $V_i(t) = e^{-L_{ii}t/C}(V_i(0) - V_{\infty}) + V_{\infty}$ is computed by matrix exponentiation and must be reproduced to 10^{-5} relative error (max norm) at $t \in \{0.25, 1, 4, 16\}\tau$. **K2 (cross-representation)**: the integrated steady-state boundary currents under $V_b = e_j$ must equal the Schur-complement column Λe_j to 10^{-6} . K2 ties the time-domain model to the static inverse problem, so the same code path serves both a solver benchmark and the measurement model of the experiment.

Table 1: Conditioning of the DtN sensitivity Jacobian, unit conductances, full boundary. $d_{\max}$: maximal graph distance to the boundary. SING.: smallest singular value at or below $\varepsilon_{\text{mach}} \sigma_{\max}$ in IEEE double precision. The last column is the 512-bit ball-arithmetic value (Sec. 2.4), given for the two unsaturated controls and for the double-precision-singular flat instances. The $\{7, 3\}$, $L = 4$ row is the preregistered confirmatory point.

Lattice	N	$ E $	$ \partial G $	$d_{\max}$	$\log_{10} \kappa$ (float64)	$\log_{10} \kappa$ (Arb)
$\{7, 3\}$, $L = 1$	35	42	28	1	1.39	
$\{7, 3\}$, $L = 2$	112	140	77	3	2.46	
$\{7, 3\}$, $L = 3$	315	399	203	5	3.27	
$\{7, 3\}$, $L = 4$	847	1078	532	7	3.89	
square, $R = 3$	29	44	20	2	1.73	
triangular, $R = 3.2$	37	87	21	2	3.42	
square, $R = 6$	113	200	44	5	5.07	
triangular, $R = 6.4$	151	401	49	5	8.67	8.67
square, $R = 10$	317	592	76	9	9.72	9.72
triangular, $R = 10.8$	421	1176	84	10	SING.	15.99
square, $R = 16$	797	1528	124	15	SING.	16.54
triangular, $R = 17.2$	1069	3071	133	16	SING.	26.95

Table 2: Probe-matched control and Neumann-to-Dirichlet (NtD) map. The $\{7, 3\}$ boundary is subsampled to the square disk’s probe count; unused boundary nodes carry zero current. Rank: exact, over $\mathbb{F}_p$ for two primes (agreeing). n_2 : unmeasured interior nodes of degree 2. σ_r/σ_{r+1} : float64 gap at the certified rank. κ_{id} : condition number on the identifiable subspace (a post hoc metric; see text). κ_{NtD} : full boundary.

Lattice	N	probes	$ E $	rank	n_2	σ_r/σ_{r+1}	$\log_{10} \kappa_{\text{id}}$	$\log_{10} \kappa_{\text{NtD}}$
$\{7, 3\}$, $L = 2$	112	44	140	117	23	1.5×10^{13}	2.70	1.99
square, $R = 6$	113	44	200	200	0	—	5.07	5.41
$\{7, 3\}$, $L = 3$	315	76	399	306	93	3.2×10^{12}	2.97	2.42
square, $R = 10$	317	76	592	592	0	—	9.72	10.08

2.8 Preregistration

Predictions for the confirmatory $\{7, 3\}$, $L = 4$ point, and for the probe-matched, Neumann-to-Dirichlet and RC analyses, were fixed in dated, version-controlled documents before the corresponding computations were run, including refutation criteria and rival functional forms [20]. Deviations are reported where they occur. The additional analyses of this version (preregistrations 5–11) were committed before each run in the same way.

3 Results

3.1 Conditioning versus network size

Table 1 and Fig. 1 give κ for all three families. On the $\{7, 3\}$ lattice the maximal boundary depth is $d_{\max} = 2L - 1$, and $\log_{10} \kappa$ rises from 1.39 ($N = 35$) to 3.89 ($N = 847$). The local exponent $d \log \kappa / d \log N$ decreases over the tested range (2.1, 1.8, 1.4), so κ grows no faster than N^2 there. On the square lattice the local exponent increases (5.7, then 10.4), $\log_{10} \kappa$ reaches 9.72 at $N = 317$, and from $N \approx 400$ onwards (triangular $N = 421$, square $N = 797$, triangular $N = 1069$) the Jacobian is numerically singular in double precision. At matched $N \approx 316$ the gap is 6.4 decades.

The preregistered confirmatory point, $\{7, 3\}$ at $L = 4$, was predicted to have $d_{\max} = 7$,

Table 3: Dimensionality in the probe-matched control. r : hyperbolic identifiable dimension. σ_1/σ_r : condition number of the square lattice’s best-conditioned r -dimensional parameter subspace (top- r right singular vectors), the most favourable r -dimensional comparison for the flat lattice.

Hyperbolic (subsamped)	r	$\log_{10} \kappa_{\text{id}}$	Flat	$ E $	$\log_{10} \kappa$ (all $ E $)	$\log_{10} \sigma_1/\sigma_r$
$\{7, 3\}$ L=2, 44 probes	117	2.70	square $R = 6$	200	5.07	2.14
$\{7, 3\}$ L=3, 76 probes	306	2.97	square $R = 10$	592	9.72	3.45

Table 4: Conditioning under inhomogeneous conductances: $\log_{10} \kappa$ of the log-parametrised Jacobian $\partial\Lambda/\partial\ln g_e$, medians over five seeds for the random regimes. The defect multiplies every edge of one interior node at maximal depth. Gap: square minus hyperbolic.

Conductances	$N \approx 112$			$N \approx 316$		
	$\{7, 3\}$ L=2	square $R=6$	gap	$\{7, 3\}$ L=3	square $R=10$	gap
unit ($g_e = 1$)	2.46	5.07	2.61	3.27	9.72	6.45
$g_e \sim \mathcal{U}[0.5, 1.5]$	2.68	5.26	2.58	3.52	9.48	5.97
$\log_{10} g_e \sim \mathcal{U}[-1, 1]$	4.34	7.91	3.57	5.19	13.45	8.26
defect $\times 100$	3.67	6.60	2.93	4.47	10.84	6.37
defect $\times 0.01$	4.00	6.65	2.65	4.80	11.14	6.34

$N \in [800, 900]$ and $\log_{10} \kappa = 4.2 \pm 0.5$; the measurements are 7, 847 and 3.89.

The first part of the mechanism is the scaling of depth with size. Boundary counts of the truncations, 28, 77, 203, 532, satisfy $b_{n+2} = 3b_{n+1} - b_n$, which grows geometrically (ratio $(3 + \sqrt{5})/2$), so the boundary is a fixed fraction of the nodes and $d_{\max} = O(\log N)$. On a flat disk $d_{\max} \sim R \sim \sqrt{N}$. In both geometries $\log \kappa$ increases with depth; on the flat lattice depth grows as a power of N . Section 3.2 shows that this is only half of the explanation: the dependence of κ on depth is itself different in the two classes.

The flat growth law beyond double precision. Version 1.0 could not say whether the flat-lattice κ grows as $e^{c\sqrt{N}}$ or as a steep power law. In ball arithmetic (Section 2.4; Table 1, last column; Fig. 1, open markers) the triangular disk at $N = 421$ has $\log_{10} \kappa = 15.99$ and the square disk at $N = 797$ has 16.54, with certified relative radii below 10^{-90} . The extrapolations fixed in the preregistration, from the unsaturated points alone, were 15.63 and 16.42 for the depth form $\log_{10} \kappa = a + c\sqrt{N}$ ($c = 0.85$ triangular, 0.64 square), against 12.5 and 12.3–13.9 for a power law. Both measurements lie within 0.4 decades of the depth form and at least 2.7 decades from the power law, and the local exponent $d \log \kappa / d \log N$ keeps increasing ($8.6 \rightarrow 16.4$ triangular, $10.4 \rightarrow 17.0$ square), as an exponential in $\sqrt{N}$ requires and a power law forbids. The largest instance, triangular $N = 1069$ (3071 unknowns), gives $\log_{10} \kappa = 26.95$ (radius 10^{-54}), one decade from the depth-form extrapolation 25.94 and eleven decades from the power-law extrapolation 15.98; the local exponent rises again, from 16.4 to 27.1. Over the whole tested range the flat-lattice condition number is therefore exponential in $\sqrt{N}$, that is, exponential in the boundary depth; limitation (i) of version 1.0 is resolved.

3.2 Other tilings: depth is not the whole mechanism

The construction of Section 3.1 extends to any $\{p, q\}$ with $(p - 2)(q - 2) > 4$. We computed κ for $\{7, 3\}$, $\{8, 3\}$, $\{5, 4\}$, $\{6, 4\}$ and $\{4, 5\}$ up to six tile layers and $N = 2888$ (Table 7), from the closed-form Gram matrix of Section 2.4 in double precision. Forming $G = J^\top J$ squares the condition number, so every value carries the bound $\varepsilon_{\text{mach}} \kappa(G)$, at most 4×10^{-9} in $\log_{10} \kappa$ here. Two controls against the direct SVD of version 1.0 pass; the second passed only after

Table 5: Dirichlet spectral gap $\lambda_{\min}(L_{ii})$ of the RC network ($C = 1$), relaxation time $\tau = C/\lambda_{\min}$ and stiffness ratio $\lambda_{\max}/\lambda_{\min}$. ^e: exploratory, not preregistered (sparse shift-invert eigensolver).

Lattice	N	$\lambda_{\min}$	τ	$\lambda_{\max}/\lambda_{\min}$
$\{7, 3\}$	35	1.0000	1.00	4.8
$\{7, 3\}$	112	0.3636	2.75	14.9
$\{7, 3\}$	315	0.2133	4.69	25.8
$\{7, 3\}$	847	0.1534	6.52	36.1
square	29	1.1716	0.85	5.8
square	113	0.2198	4.55	35.4
square	317	0.0688	14.54	115.3
square	797	0.0256	39.06	311.5
triangular	37	1.3258	0.75	6.3
triangular	151	0.2703	3.70	32.8
triangular	475	0.0768	13.02	116.7
triangular	1213	0.0281	35.56	319.6
$\{7, 3\}^e$	2240	0.1232	8.11	—
$\{7, 3\}^e$	5887	0.1058	9.45	—

Table 6: Integrator controls for the RC network ($\text{rtol} = 10^{-8}$, $\text{atol} = 10^{-10}$). K1: max-norm relative error against the matrix-exponential solution at $t \in \{0.25, 1, 4, 16\}\tau$, threshold 10^{-5} . K2: relative error of the integrated steady-state boundary currents against the Schur-complement column Λe_j , threshold 10^{-6} .

Network	Integrator	K1 error		K2 error	
$\{7, 3\}$, $L = 2$	SciPy BDF	7.3×10^{-9}	pass	4.2×10^{-13}	pass
$\{7, 3\}$, $L = 2$	rusty-SUNDIALS CVODE (BDF)	1.1×10^{-8}	pass	1.1×10^{-12}	pass
square, $R = 6$	SciPy BDF	1.4×10^{-8}	pass	4.1×10^{-13}	pass
square, $R = 6$	rusty-SUNDIALS CVODE (BDF)	2.6×10^{-8}	pass	1.1×10^{-12}	pass

the control tolerance was amended, before any tiling had been computed, to include this floor (preregistration 11, Deviation 1). Every interior node of every tiling has degree q .

Five preregistered predictions were tested: four held and one was refuted. $\log \kappa$ is concave in the number of layers for $\{7, 3\}$ and $\{8, 3\}$, and $\{8, 3\}$ is better conditioned than $\{7, 3\}$ at equal layer count ($2.31 < 2.46$, $3.06 < 3.27$, $3.69 < 3.89$). The prediction that the local exponent $d \log \kappa / d \log N$ between the two largest sizes is at most 1.5 for $\{8, 3\}$ and at most 1.0 for the $q = 4, 5$ tilings was refuted for two of the three: the exponents are 1.10 ($\{8, 3\}$), 0.98 ($\{6, 4\}$), 1.19 ($\{5, 4\}$) and 1.06 ($\{4, 5\}$). They decrease with size (for $\{5, 4\}$: 1.50, 1.35, 1.19) but stay above one in the tested range.

The main result concerns depth. At equal maximal depth $d_{\max}$ the hyperbolic tilings agree to within 0.45, 0.25, 0.56 and 0.21 decades at $d_{\max} = 1, 2, 3, 5$, although N varies by a factor of about 15 at fixed depth (at $d_{\max} = 3$, N runs from 112 to 1710). Within the hyperbolic class depth is therefore nearly a sufficient statistic for κ . Across classes it is not: at $d_{\max} = 5$ the square (5.07) and triangular (8.67) lattices exceed the hyperbolic maximum (3.27) by at least 1.8 decades (Fig. 3). The figure shows why. The increment of $\log_{10} \kappa$ per unit of depth falls from about 0.66 ($d_{\max}$ from 1 to 2, $\{5, 4\}$) to 0.57 ($2 \rightarrow 3$) and, on $\{7, 3\}$, to 0.40 ($3 \rightarrow 5$) and 0.31 ($5 \rightarrow 7$), whereas for the flat lattices it is roughly constant, 1.1–1.2 per unit. The scaling of depth with size, discussed above, is thus true but only half of the explanation; the other half is the different dependence of κ on depth. We have a derivation of neither dependence.

Table 7: $\log_{10} \kappa$ of the DtN sensitivity Jacobian (maximal depth $d_{\max}$ in brackets) for five hyperbolic tilings, unit conductances, full boundary; L is the number of tile layers. Closed-form Gram method, double precision, error bound at most 4×10^{-9} in $\log_{10} \kappa$.

Tiling	$L=1$	$L=2$	$L=3$	$L=4$	$L=5$	$L=6$	$N_{\max}$
$\{7, 3\}$	1.39 (1)	2.46 (3)	3.27 (5)	3.89 (7)			847
$\{8, 3\}$	1.35 (1)	2.31 (3)	3.06 (5)	3.69 (7)			2888
$\{5, 4\}$	0.00 (0)	1.08 (1)	1.75 (2)	2.32 (3)	2.82 (4)		1160
$\{6, 4\}$	0.00 (0)	1.11 (1)	1.78 (2)	2.35 (3)			1710
$\{4, 5\}$	0.00 (0)	1.10 (1)	1.54 (1)	1.99 (2)	2.48 (3)	2.87 (3)	1008

Table 8: Localisation of a single-node defect at the deepest node, matched-filter decoder: ε_{loc} is the largest measurement-noise level (relative, i.i.d. Gaussian) at which top-1 accuracy stays at least 0.9, for contrast $f = 2$ and $f = 1.25$; the margin is $\varepsilon_{\text{loc}}/(3 \times 10^{-4})$, the paper’s precision budget. Twenty trials per cell, half-decade noise grid; ideal board, known contrast set.

Lattice	N	$\varepsilon_{\text{loc}} (f=2)$	margin	$\varepsilon_{\text{loc}} (f=1.25)$
$\{7, 3\}, L = 2$	112	1×10^{-1}	333	3×10^{-2}
square, $R = 6$	113	1×10^{-2}	33	3×10^{-3}
$\{7, 3\}, L = 3$	315	$> 3 \times 10^{-1}$	$> 10^3$	1×10^{-1}
square, $R = 10$	317	3×10^{-3}	10	1×10^{-3}

3.3 Identifiability is exact; the difference is conditioning

Exact ranks over $\mathbb{F}_p$ show that $\{7, 3\}$ at $L = 1, 2$, the square disk at $N = 113$, and triangular disks at $N = 187$ and $N = 475$ all have full column rank, certified for both primes. The last is comparable in size to flat instances whose floating-point SVD is at the precision floor. At the sizes tested, every edge is algebraically recoverable on both geometries; what separates them is the numerical precision needed to exploit that recoverability. This is consistent with the criticality theory of circular planar networks [6, 8], which concerns identifiability and is silent about conditioning.

3.4 Probe-matched control

A truncated hyperbolic tiling has more boundary probes than a flat disk of equal N (77 versus 44 at $N \approx 112$; 203 versus 76 at $N \approx 316$). To test whether the advantage is only a probe-count effect, the $\{7, 3\}$ boundary was subsampled to the flat probe count, evenly in angle, with unused boundary nodes carrying zero current.

The preregistered comparison of full-Jacobian condition numbers is undefined in this design: every subsampled Jacobian is singular. The exact ranks explain why. Each unused boundary node of degree two becomes an unmeasured interior node joining two conductances in series, of which only the series combination $g_1 g_2 / (g_1 + g_2)$ affects the boundary data. This is the standard non-recoverable series reduction of circular planar networks [7], and the exact deficiency equals the number of such nodes (23 at $L = 2$, 93 at $L = 3$; Table 2). The floating-point spectrum shows a gap of 10^{12} – 10^{13} at exactly the certified rank, so the float SVD resolves the rank unambiguously.

As a replacement metric, chosen after the singularity was observed and therefore post hoc, we report κ on the identifiable subspace, discarding exactly the certified number of singular values. It is $10^{2.70}$ versus $10^{5.07}$ at $N \approx 112$ and $10^{2.97}$ versus $10^{9.72}$ at $N \approx 316$. The two sides then recover different targets: 306 identifiable combinations of 399 hyperbolic conductances, versus all 592 square-lattice conductances. Recovering fewer combinations is a genuine loss of resolution, not a bookkeeping choice.

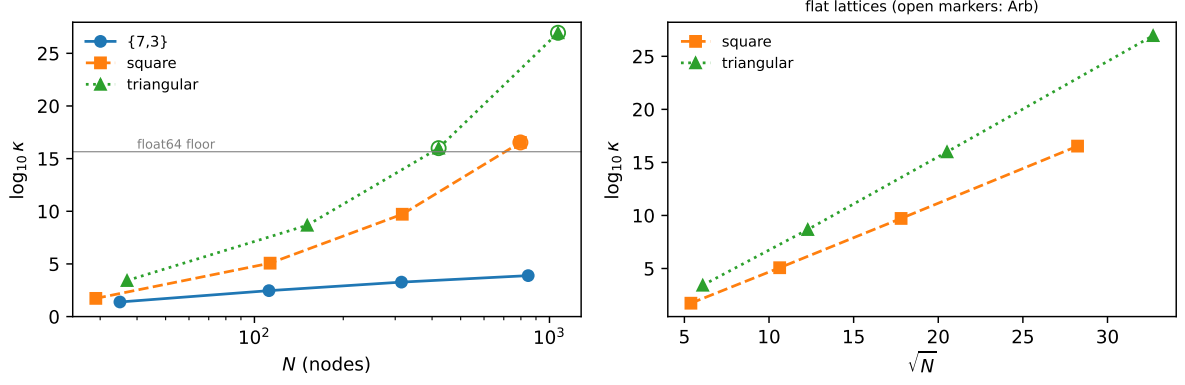


Figure 1: Left: $\log_{10} \kappa$ of the DtN sensitivity Jacobian versus N . Filled markers are double-precision values; open markers are the ball-arithmetic values of Section 2.4 for instances that are singular in double precision (crosses at the floor where no such value is available). Right: the two flat families against $\sqrt{N}$, the abscissa on which the depth mechanism predicts a straight line.

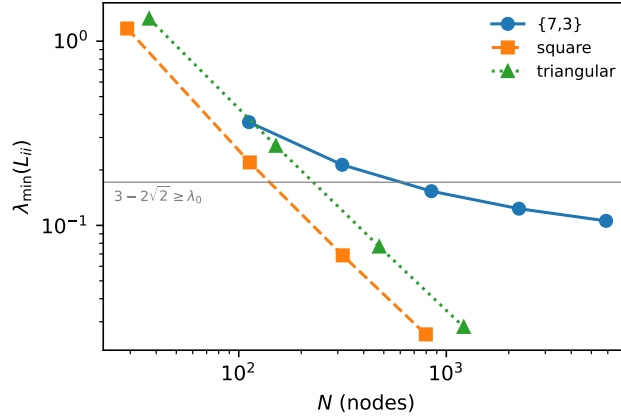


Figure 2: Dirichlet spectral gap $\lambda_{\min}(L_{ii})$ of the RC network; the $\{7, 3\}$ points at $N = 2240$ and 5887 are exploratory. The grey line is the upper bound $3 - 2\sqrt{2}$ on the bottom of the spectrum of the infinite $\{7, 3\}$ graph.

Version 1.0 of this paper read the 6.7-decade gap as evidence that the advantage survives probe matching. That reading does not survive a dimension-matched comparison, which we preregistered in response to review and report in Table 3. For the square lattice at the same probe count we take the *best*-conditioned r -dimensional parameter subspace, spanned by its top- r right singular vectors, with r equal to the hyperbolic identifiable dimension; its condition number is σ_1/σ_r , the most favourable r -dimensional comparison the flat lattice can be given. At $N \approx 112$ ($r = 117$) the flat subspace is *better* conditioned, $10^{2.14}$ against $10^{2.70}$; at $N \approx 316$ ($r = 306$) the hyperbolic side is better by 0.5 decades, $10^{2.97}$ against $10^{3.45}$, not by the 1–4 decades we had predicted. Most of the 6.7-decade gap is therefore dimensionality, and at matched probe count *and* matched dimension the residual advantage is at most half a decade at these sizes. The full-boundary comparison of Section 3.1 is unaffected: a boundary that holds a fixed fraction of the nodes is the geometric property at issue, and subsampling it discards exactly that property. What is withdrawn is the claim that the advantage is independent of probe count.

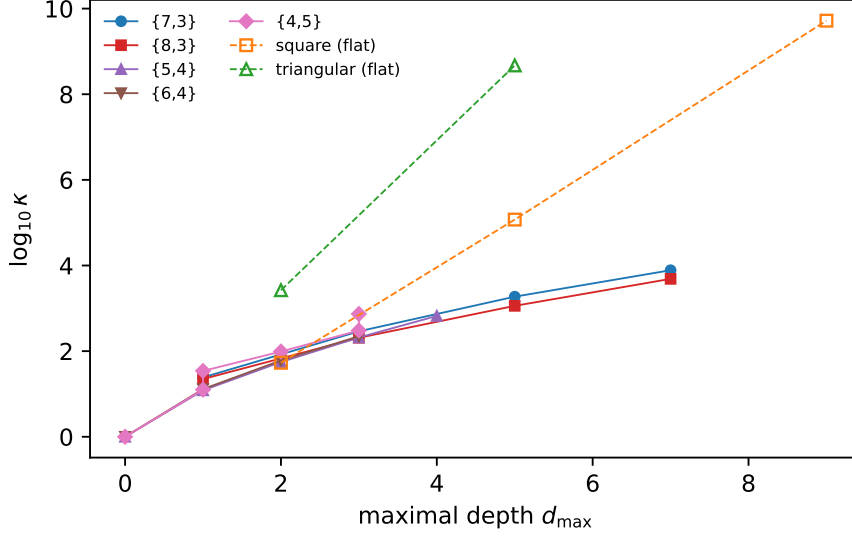


Figure 3: $\log_{10} \kappa$ against maximal depth $d_{\max}$: five hyperbolic tilings (filled markers) nearly coincide and are concave in depth; the square and triangular lattices (open markers, double-precision values) are roughly linear and lie far above at equal depth.

3.5 Neumann-to-Dirichlet map

For the map measured by current-injection hardware, the ordering is preserved, as preregistered: $\log_{10} \kappa_{\text{NtD}} = 1.99$ and 2.42 ($\{7, 3\}$, $L = 2, 3$) versus 5.41 and 10.08 (square, $R = 6, 10$) (Table 2).

3.6 Inhomogeneous conductances

Version 1.0 evaluated κ only at unit conductances. In response to review we preregistered and ran three inhomogeneous regimes (Table 4), reporting the condition number of the log-parametrised Jacobian $\partial \Lambda / \partial \ln g_e = g_e \partial \Lambda / \partial g_e$, that is, for relative perturbations, the quantity that component tolerances set. Under mild disorder $g_e \sim \mathcal{U}[0.5, 1.5]$ the median gap at $N \approx 316$ is 6.0 decades (unit conductances: 6.4) and the hyperbolic κ moves by 0.25 decades, as predicted (the refutation threshold was a gap below 3). Under two-decade log-uniform disorder both condition numbers rise, the hyperbolic by 1.9 decades and the square by 3.7, so the gap *widens* to 8.3 decades (3.6 at $N \approx 112$); the prediction was a gap of at least 3. A $\times 100$ or $\times 0.01$ contrast on every edge of one node at maximal depth raises the hyperbolic κ by 1.2 and 1.5 decades respectively (prediction: below 2) and leaves the gap at 6.3–6.4 decades. All three predictions held: the advantage is a property of the geometry, not of the homogeneous background.

3.7 Defect detection and localisation in simulation

A board would be used to detect and locate faults, so we tested this in simulation, preregistering each step (preregistrations 5–10) and keeping refutations. A defect multiplies every edge of one interior node by a factor f ; the response saturates (at the deepest node $f = 2$ gives 27–37% of the $f = 100$ signal). The data are Neumann-to-Dirichlet maps with independent i.i.d. Gaussian noise ε , relative to their rms entry; the precision budget of Section 3.6 is $\varepsilon = 3 \times 10^{-4}$.

Detection. Against a known baseline a deep $\times 100$ defect is detected at $\varepsilon = 3 \times 10^{-4}$ on all four lattices, and up to $\varepsilon = 10^{-1}$ on $\{7, 3\}$ $L = 3$ against 10^{-2} on square $R = 10$. Compared with the ideal simulation, in the presence of component tolerance $g_e = 1 + \tau u_e$, $u_e \sim \mathcal{U}[-1, 1]$, it is detected up to $\tau = 5\%$ on both hyperbolic lattices and up to 1% on square $R = 10$; comparing a board with its own earlier measurement it is detected at every τ tested. A topological readout is

less sensitive than the plain resistance metric: the persistent homology (Gudhi) of the boundary effective-resistance metric shows no signature of a deep defect above a disorder null, and its noise tolerance is at least 3.3 times lower than that of the metric.

Localisation. A matched-filter decoder over all interior nodes and a known set of eight contrasts recovers the defect node with top-1 accuracy 1.00 in all 60 preregistered cells at $\varepsilon = 3 \times 10^{-4}$ (contrasts 0.8 to 100, three depths, four lattices), on flat and hyperbolic lattices alike. We had predicted that the flat lattices would fail at deep nodes and weak contrast; two of three predictions were refuted. A noise sweep (Table 8) locates the failure boundary: at the deepest node the noise tolerance of the hyperbolic lattice exceeds that of the flat lattice by at least $100\times$ at $N \approx 316$, and at the 3×10^{-4} budget the flat board has only $3\text{--}10\times$ headroom. Component tolerance up to 5% (a fresh random board per trial, ideal-model decoder, both maps from the same board) does not degrade localisation: 45 of 48 cells are perfect, and the three that are not belong to the noise-limited cell of square $R = 10$ (0.55–0.60 against 0.50 on an ideal board). We had predicted otherwise, carrying over the model-based detection result to a differential measurement.

What this does not show. The defects are single nodes; the decoder is an oracle with a known contrast set; the noise is i.i.d.; the cells use twenty trials and a half-decade noise grid; the deepest class is a single node on the square lattices; and everything is simulation. The advantage of the hyperbolic lattice is a larger noise margin, not that a flat board cannot localise a defect at ideal noise. Why the flat lattice fails at noise $100\times$ lower, far more than the $8\text{--}10\times$ ratio of the boundary signals, is not explained.

3.8 RC dynamics and the Dirichlet spectral gap

The preregistered prediction was that $\lambda_{\min}(L_{ii})$ on $\{7, 3\}$ is constant over $L = 2\text{--}4$, refuted if it fell by a factor of two or more. It fell by 2.37 ($0.364 \rightarrow 0.153$), so the prediction is refuted as stated. The flat-lattice prediction, $\lambda_{\min} \sim 1/N$, holds (Table 5, Fig. 2).

The asymptotic question is nevertheless settled by the following observation.

Proposition 1. *Suppose every interior node of a finite truncation G_L of the $\{7, 3\}$ tiling has degree 3. Then $\lambda_{\min}(L_{ii}(G_L)) \geq \lambda_0$, the bottom of the spectrum of the combinatorial Laplacian of the infinite $\{7, 3\}$ graph, and $\lambda_0 > 0$. If moreover the interiors are nested and exhaust the tiling, then $\lambda_{\min}(L_{ii}(G_L))$ is non-increasing in L and converges to λ_0 .*

Proof sketch. If interior nodes have full degree, then for f supported on the interior, $f^\top L_{ii} f = \sum_{\text{edges}} (f_a - f_b)^2$ with the sum over all edges of the infinite graph, so L_{ii} is the compression of the infinite Laplacian to the interior. By the Rayleigh quotient, $\lambda_{\min}(L_{ii})$ is the infimum of that quotient over the interior, which is at least λ_0 , the infimum over all finitely supported functions. Nesting gives monotonicity, and exhaustion gives convergence, since every finitely supported function is eventually supported in some interior. Finally $\lambda_0 > 0$ because the $\{7, 3\}$ graph has a positive edge-isoperimetric constant [21] and bounded degree [22, 23]. $\square$

For $L = 1, \dots, 6$ we verified that every interior node has degree three (integer degree counts), and for $L = 2, \dots, 6$ that the interior of G_L is G_{L-1} : interior node coordinates match those of G_{L-1} to 10^{-9} in the Poincaré disk, and the induced edge set equals the edge set of G_{L-1} (interior counts 7, 35, 112, 315, 847, 2240). For general L the degree-three hypothesis follows from two facts. First, the union of the heptagons within L dual steps of the central tile is a closed topological disk: it is a connected union of closed tiles in the plane, and its Euler characteristic $V - E + F = 1$, which the network builder checks for every instance, forces genus zero with one boundary component. Second, every vertex of the tiling lies on exactly three heptagons whose union covers a neighbourhood of it. If all three are in G_L , the vertex's three edges are in G_L (each edge is shared by two of the heptagons) and the vertex is not on the outer face. If one is missing, the missing sector belongs to the complement of the disk, which is connected

and hence part of the outer face, so the vertex is a boundary node. Thus an interior node has all three heptagons, and degree three. Nesting is immediate: a vertex whose three heptagons lie within L layers has them within $L + 1$, so $\text{int } G_L \subset \text{int } G_{L+1}$, and the interiors exhaust the tiling. The stronger statement $\text{int } G_L = G_{L-1}$ is not needed by the proposition and remains certified only for $L \leq 6$. The computed values 0.364, 0.213, 0.153, 0.123, 0.106 ($L = 2-6$; the last two exploratory) decrease with successive ratios 0.59, 0.72, 0.80, 0.86, as the proposition requires, and lie below the tree bound $\lambda_0 \leq 3 - 2\sqrt{2} \approx 0.172$ from $L = 4$ on. The finite-size refutation and the uniform bound are compatible: the refuted prediction concerned the rate of approach to λ_0 , not its existence.

Physically, the slowest RC relaxation time of a hyperbolic network is bounded, $\tau \leq C/\lambda_0$, for every N , whereas on flat disks τ grows linearly in N . At $N \approx 800$, $\tau = 6.5C$ versus $39C$, and the stiffness ratio is 36 versus 311.

3.9 Integrator controls

Both controls pass for both integrators on $\{7, 3\}$, $L = 2$ and square $R = 6$ (Table 6). K1 errors are of order 10^{-8} for both, well inside the 10^{-5} threshold, so the two codes agree with the exact solution, and hence with each other, to about 3×10^{-8} . K2 errors are of order 10^{-13} (SciPy) and 10^{-12} (CVODE). The CVODE leg used the rusty-SUNDIALS Python binding (version 6.0.0) built from commit `af4886f`. The same harness, packaged as a benchmark with the two fixture networks, has been contributed to rusty-SUNDIALS as an example and regression test (`examples/python/rc_network`, pull request #62, merged as commit `4ce8abb`).

4 Discussion and limitations

The central result is geometric and simple: an inverse problem whose information decays with distance from the boundary is conditioned by the depth distribution of its unknowns, and hyperbolic truncations keep every unknown within $O(\log N)$ of the boundary. Negative curvature buys a better-conditioned discrete Calderón problem, and for the RC model a uniformly bounded relaxation time. Depth is the controlling variable within the hyperbolic class; across geometries the dependence of κ on depth differs (concave versus linear), so depth scaling explains the gap only together with that difference (Section 3.2).

The limitations are as follows. (i) The flat-lattice growth law $\kappa \sim e^{c\sqrt{N}}$ is supported by the ball-arithmetic values at the sizes tested, not proven; the constant c is fitted, not derived. (ii) κ is a linearised sensitivity; disorder and single defects are tested (Section 3.6), but no nonlinear reconstruction algorithm is benchmarked. (iii) Exact rank certificates cover $N \leq 475$ for flat lattices; full rank there does not exclude deficiency at larger N . (iv) The probe-matched comparison uses a post hoc metric; once dimension is matched as well, the residual advantage is at most half a decade, and reversed at $N \approx 112$ (Section 3.4). (v) Conditioning is computed for five hyperbolic tilings up to six layers and $N = 2888$, with unit conductances and full boundary; disorder and probe subsampling are studied only on $\{7, 3\}$ and the flat lattices, and exact rank certificates cover only $\{7, 3\}$ and the flat lattices. (vi) The two-integrator cross-validation covers two networks of about 110 nodes; larger and stiffer instances are not yet cross-validated. (vii) The novelty search was targeted, not systematic. (viii) Detection and localisation are simulations of single-node defects with an oracle decoder, i.i.d. noise, tolerance up to 5% and at most twenty trials per cell; multi-node defects, unknown contrasts, larger tolerance and hardware noise are untested.

5 Research directions

Geometry-controlled stiffness benchmark for rusty-SUNDIALS. Equation (4) with the K1/K2 controls is a stiff linear system with an exact solution and a static cross-check, at tunable size and stiffness. Its stiffness is bounded in N on hyperbolic networks and grows linearly on flat ones. The two-network version is now part of rusty-SUNDIALS as a regression benchmark. Natural extensions are:

- cross-validation at $N \sim 10^3\text{--}10^4$, where the flat stiffness exceeds 300;
- an analytic-Jacobian argument ($-L_{ii}/C$) in the Python binding, which it does not currently accept, so that solver cost can be measured as a function of geometry;
- sparse linear algebra inside the BDF Newton iterations.

Difference imaging. Single-node localisation from the difference of two boundary maps is now tested in simulation (Section 3.7). Open are multi-node and extended defects, unknown contrasts, tolerance above 5%, hardware noise, and an explanation of why the flat lattice fails at noise $100\times$ below the hyperbolic one.

Other tilings and a scaling law. Section 3.2 covers five tilings. Open are a derivation of the concave (hyperbolic) and linear (flat) dependence of κ on depth, disorder, probe subsampling and exact rank certificates on the other tilings, tilings with $q \geq 6$, and a test of whether the exponents of the $q = 4, 5$ tilings keep falling at larger N .

Tree metrics. Gromov-hyperbolic graphs are quasi-trees, and effective resistance on trees is additive, which links boundary resistance data to tree-metric and circular split-system reconstruction [24]. A stability theory of that reconstruction might explain the polynomial conditioning found here.

Tabletop measurement. A printed-circuit $\{7, 3\}$ network at $L = 2$ ($N = 112, 140$ resistors), with a matched flat control, can be probed in the time domain through eq. (4). The predicted observables are the relaxation time, the stiffness, and the conditioning of the measured Neumann-to-Dirichlet map. Since $\kappa \approx 10^{2.5}$ at this size, a linearised static reconstruction to 10% requires a total relative measurement error of about 3×10^{-4} . That precision budget, rather than component tolerance alone, determines whether the static comparison, or only the time-domain one, is feasible at $L = 2$. In simulation (Section 3.7), a single-node fault is localised at this budget on either geometry, and component tolerance up to 5% does not matter for a same-board differential measurement; the benefit of the hyperbolic layout is noise robustness, and the board’s role is to test that mechanism and the relaxation time.

Formalisation. The proposition above, and the finite-dimensional core of the bulk-boundary correspondence for the Su–Schrieffer–Heeger chain, are candidates for machine-checked proofs in Lean 4 with Mathlib, whose Schur-complement library already contains the block-matrix identities used here.

Changes from version 1.0

Version 1.1 responds to a peer review of version 1.0 (both the review and the response are in the repository). Every added computation was preregistered before it was run. (a) The flat-lattice growth law is now measured beyond double precision in ball arithmetic (Section 2.4, Section 3.1). (b) Conditioning under disorder and defects is added (Section 3.6); all predictions

held. (c) A dimension-matched probe control refuted our own prediction and *corrects* version 1.0: at matched probe count and matched dimension the advantage is at most half a decade (Section 3.4). (d) The degree-three premise of Proposition 1 is now argued for general L (Section 3.8). Version 1.0 remains available under its own DOI; the ledger records the correction against the original claim without altering it.

Changes from version 1.1

Version 1.2 (draft) adds the following, all preregistered before it was run; nothing in versions 1.0 or 1.1 is altered. (a) Conditioning across five hyperbolic tilings and a *qualification* of the mechanism statement of version 1.1: depth scaling is only half of the explanation (Section 3.2). One prediction of that analysis was refuted, and a control tolerance was amended before any result was seen. (b) Defect detection and localisation in simulation, against measurement noise and component tolerance (Section 3.7); two localisation predictions and one tolerance prediction were refuted. (c) The verification table below now also lists the claims of version 1.1 that it omitted. (d) Erratum: in version 1.1, Section 3.1, a space is missing after “27.1” (a typesetting slip; no content is affected).

Verification status

Every claim in this paper is recorded in a machine-readable ledger [20] with an evidence tier: X (numerical), C (argument), L (literature), B (exact computation), and a SHA-256 digest of its evidence. For every claim in the table below, the ledger gate verifies the digest against the released evidence store and reports no findings.

Claim	Tier	Ledger
Conditioning versus N (Table 1), preregistered $L = 4$ point	X	H0-X-0002
Full rank of all tested full-boundary networks	B	H0-B-0001
Probe-matched deficiency = number of unmeasured degree-2 nodes	B	H0-B-0002
Probe-matched κ on identifiable subspace (post hoc)	X	H0-X-0003
Neumann-to-Dirichlet ordering	X	H0-X-0004
H2 finite-size prediction refuted; flat $\lambda_{\min} \sim 1/N$	X	H2-X-0001
$\lambda_{\min}$ at $L = 5, 6$ (exploratory)	X	H2-X-0002
Interior degree 3, nested interiors, $L \leq 6$	B	H2-B-0001
$\lambda_0 > 0$ for $\{7, 3\}$	L	H2-L-0001
Uniform bound $\tau \leq C/\lambda_0$ (Proposition)	C	H2-C-0001
Integrator controls K1/K2, SciPy and rusty-SUNDIALS CVODE	X	H2-X-0004
Dimension-matched probe control (corrects version 1.0)	X	H0-X-0005
Conditioning under disorder and defects	X	H0-X-0006
Flat growth law beyond double precision (ball arithmetic)	X	H0-X-0007
Conditioning across five tilings (depth nearly universal within the class)	X	H0-X-0008
No persistent-homology signature of a deep defect	X	H3-X-0001
Defect detection against measurement noise	X	H3-X-0002
Defect detection against component tolerance	X	H3-X-0003
Defect response saturates in the contrast	X	H3-X-0004
Localisation at the 3×10^{-4} budget	X	H3-X-0005
Localisation noise margin	X	H3-X-0006
Localisation under component tolerance	X	H3-X-0007

Data and code availability

Code, preregistrations, raw data (JSON), network edge lists, Dirichlet-to-Neumann matrices, the exact rank certificates (`data/exact_rank_full.json`, `data/identifiability.json`) and

the claim ledger with its evidence store are available at [20] and in the archived dataset accompanying this preprint. Every table in this paper is generated from the released data by `paper/make_assets.py`.

Use of AI tools

Code, numerical experiments and manuscript drafting were carried out with the assistance of Claude (Anthropic) as a research and programming assistant. The author reviewed all claims against their evidence and is responsible for the content.

References

- [1] C. L. Kane and E. J. Mele. Z_2 topological order and the quantum spin Hall effect. *Phys. Rev. Lett.*, 95:146802, 2005.
- [2] M. Z. Hasan and C. L. Kane. Colloquium: Topological insulators. *Rev. Mod. Phys.*, 82:3045, 2010.
- [3] A. P. Schnyder, S. Ryu, A. Furusaki, and A. W. W. Ludwig. Classification of topological insulators and superconductors in three spatial dimensions. *Phys. Rev. B*, 78:195125, 2008.
- [4] J. Maldacena. The large N limit of superconformal field theories and supergravity. *Adv. Theor. Math. Phys.*, 2:231–252, 1998.
- [5] S. Ryu and T. Takayanagi. Holographic derivation of entanglement entropy from the anti-de Sitter space/conformal field theory correspondence. *Phys. Rev. Lett.*, 96:181602, 2006.
- [6] E. B. Curtis, D. Ingerman, and J. A. Morrow. Circular planar graphs and resistor networks. *Linear Algebra Appl.*, 283:115–150, 1998. doi: 10.1016/S0024-3795(98)10087-3.
- [7] E. B. Curtis and J. A. Morrow. *Inverse Problems for Electrical Networks*, volume 13 of *Series on Applied Mathematics*. World Scientific, Singapore, 2000.
- [8] Y. Colin de Verdière. Réseaux électriques planaires I. *Comment. Math. Helv.*, 69:351–374, 1994. doi: 10.1007/BF02564493.
- [9] L. Borcea, V. Druskin, F. Guevara Vasquez, and A. V. Mamonov. Resistor network approaches to electrical impedance tomography. In *Inside Out II*, volume 60 of *MSRI Publications*. Cambridge University Press, 2012.
- [10] L. Borcea, V. Druskin, A. V. Mamonov, and F. Guevara Vasquez. Circular resistor networks for electrical impedance tomography with partial boundary measurements. *Inverse Problems*, 26:045010, 2010. doi: 10.1088/0266-5611/26/4/045010.
- [11] L. Borcea, V. Druskin, A. V. Mamonov, and F. Guevara Vasquez. Pyramidal resistor networks for electrical impedance tomography with partial boundary measurements. *Inverse Problems*, 26:105009, 2010. doi: 10.1088/0266-5611/26/10/105009.
- [12] S. Ervedoza and F. de Gournay. Uniform stability estimates for the discrete Calderón problems. *Inverse Problems*, 27:125012, 2011.
- [13] A. J. Kollár, M. Fitzpatrick, and A. A. Houck. Hyperbolic lattices in circuit quantum electrodynamics. *Nature*, 571:45–50, 2019.
- [14] P. M. Lenggenhager et al. Simulating hyperbolic space on a circuit board. *Nat. Commun.*, 13:4373, 2022.

- [15] M. Asaduzzaman, S. Catterall, J. Hubisz, R. Nelson, and J. Unmuth-Yockey. Holography on tessellations of hyperbolic space. *Phys. Rev. D*, 102:034511, 2020.
- [16] P. Basteiro, G. Di Giulio, J. Erdmenger, J. Karl, R. Meyer, and Z.-Y. Xian. Towards explicit discrete holography: Aperiodic spin chains from hyperbolic tilings. *SciPost Phys.*, 13:103, 2022.
- [17] P. Virtanen et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nat. Methods*, 17:261–272, 2020.
- [18] X. Callens. rusty-SUNDIALS: a Rust implementation of SUNDIALS solvers with Python bindings. Software, <https://github.com/xaviercallens/rusty-SUNDIALS>, commit af4886f, 2026.
- [19] A. C. Hindmarsh, P. N. Brown, K. E. Grant, S. L. Lee, R. Serban, D. E. Shumaker, and C. S. Woodward. SUNDIALS: Suite of nonlinear and differential/algebraic equation solvers. *ACM Trans. Math. Softw.*, 31(3):363–396, 2005.
- [20] X. Callens. Track H: hyperbolic resistor-network conditioning – code, data, preregistrations and claim ledger. <https://github.com/xaviercallens/SocrateAI-Scientific-CondensedMatterTheory>, directory experiments/track_h_hyperbolic_network, 2026.
- [21] O. Häggström, J. Jonasson, and R. Lyons. Explicit isoperimetric constants and phase transitions in the random-cluster model. *Ann. Probab.*, 30(1):443–473, 2002.
- [22] J. Dodziuk. Difference equations, isoperimetric inequality and transience of certain random walks. *Trans. Amer. Math. Soc.*, 284:787–794, 1984.
- [23] B. Mohar. Isoperimetric inequalities, growth, and the spectrum of graphs. *Linear Algebra Appl.*, 103:119–131, 1988. doi: 10.1016/0024-3795(88)90224-8.
- [24] S. Forcey and D. Scalzo. Phylogenetic networks as circuits with resistance distance. *Front. Genet.*, 11:586664, 2020.